Characters

a list of characters
who appear significantly
at several different places in the book

NOTE: The following descriptions are meant to serve solely as reminders of and cross-references to each person's various appearances in this book. These descriptions are *not* intended as biographical sketches. (Most of these people have made major contributions to science that are not relevant to this book and therefore are not listed here.) The principal criterion for inclusion in this section is *not* importance of contributions, but rather multiple appearances of the person at several different locations in the book.

Baade, WaIter (1893–1960). German born, American optical astronomer; with Zwicky, developed the concept of a supernova and its connection to neutron stars (Chapter 5); identified the galaxies associated with cosmic radio sources (Chapter 9).

Bardeen, James Maxwell (b. 1939). American theoretical physicist; showed that many or most black holes in our Universe should be rapidly spinning and, with Petterson, predicted the influence of the holes' spins on surrounding accretion disks (Chapter 9); with Carter and Hawking, discovered the four laws of black-hole mechanics (the laws of evolution of black holes) (Chapter 12).

Bekenstein, Jacob (b. 1947). Israeli theoretical physicist; student of Wheeler's; with Hartle, showed that one cannot discern, by any external study of a black hole, what kinds of particles were among the material that formed it (Chapter 7); proposed that the surface area of a black hole is its